YOLISA HLOHLA

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PROFESSIONAL SUMMARY

Computer Engineering graduate with strong foundations in programming, operating systems, and hardware-software integration. Concentrated on in scalable software systems, network automation, web development and performance optimization. Proven ability to deliver production-ready solutions that improve operational efficiency, enhance system reliability, and accelerate incident response. Experienced across the full development lifecycle, from system architecture and algorithm design to deployment, testing, and monitoring.

CORE SKILLS

Languages: Python, C/C++, Java, JavaScript, SQL, Bash
Web & Backend: Django, Flask, Node.js, React, REST APIs
Networking & SDN: YANG, NETCONF/RESTCONF, OpenDaylight, BGP, OSPF, Cisco IOS
Data & Optimization: Linear Programming, NetworkX, PuLP, Data Visualization
DevOps & Systems: Linux, Docker, Git, CI/CD, Prometheus, Grafana, SNMP, Wireshark
Databases: PostgreSQL, MongoDB, MySQL
Embedded & IoT: Arduino, Raspberry Pi, Embedded C

TECHNICAL EXPERIENCE

Student Researcher & Developer | Academic Labs & Engineering Projects *2023 – 2025*

- Completed hands-on labs involving amplifiers, multiplexers, counters, and shift-register circuits.
- Applied Kirchhoff's laws to analyse and verify electrical circuit behaviour.
- Performed optics experiments including lenses, diffraction, interference, and polarization.
- Conducted Doppler-effect measurements and wave-frequency analysis in physics labs.
- Automated system monitoring and log analysis pipelines, reducing incident response time by ~60%.
- Built Bash and Python tooling that cut manual system administration work by 50%.
- Diagnosed and resolved cross-layer issues (software, OS, network, hardware) to sustain 99% lab availability.
- Translated engineering requirements into tested, documented solutions across small cross-functional teams.

SELECTED PROJECTS

SDN Network Automation Platform - *Python · YANG · NETCONF/RESTCONF · OpenDaylight*

- Reduced manual network configuration effort by ~80% by building an end-to-end automated provisioning system across multi-vendor environments.
- Designed YANG data models and implemented rollback/validation logic enabling zero-downtime updates in simulated production conditions.
- Delivered real-time telemetry dashboards (Prometheus/Grafana) to surface misconfigurations and performance regressions proactively.

Network Traffic Engineering Optimization - *Python · Linear Programming · NetworkX · PuLP*

- Improved link utilization by ~35% by developing LP optimization models that directly reduced simulated congestion and infrastructure cost.
- Translated mathematical models into production-ready Python pipelines and generated visual reports to support capacity planning decisions.

Secure Banking API Platform - *Python · Django · MongoDB · JWT · RBAC*

- Engineered a modular backend handling 1,000+ concurrent requests with JWT authentication, RBAC, encryption, and financial-grade access controls.
- Achieved 95% automated test coverage, accelerating feature iteration while maintaining reliability.

Broadcasting & Audio Systems - Electronic Communications 3 Research

- Investigated audio system architecture and broadcasting principles for the GA6 Electronic Communications project.

- Researched signal-processing methods used in audio transmission and reception.
- Examined amplification, modulation techniques, and frequency allocation in radio and audio broadcast systems.
- Delivered a comprehensive research report covering audio system components and broadcasting infrastructure.

Networking Labs - Cisco Packet Tracer, VyOS, Routing Protocols

- Configured routers and switches using Cisco Packet Tracer and VyOS for academic networking labs.
- Practiced troubleshooting, subnetting, and IP addressing across multi-layer network topologies.
- Built networks from scratch implementing RIP, OSPF, EIGRP, and VLAN segmentation.
- Gained hands-on experience with network security controls including ACLs and basic firewall rules.

Cybersecurity Learning Lab - VirtualBox, Wireshark, Security Research

- Set up a personal cybersecurity lab using VirtualBox virtual machines for practical security experimentation.
- Used Wireshark for packet capture, protocol analysis, and traffic inspection.
- Conducted research on load-balancing mechanisms from a cybersecurity perspective, identifying potential vulnerabilities.
- Explored penetration-testing concepts and system-hardening techniques.

GNS3 Advanced Networking Labs - Network Systems 3

- Built advanced network simulations using the GNS3 emulation platform beyond basic Packet Tracer capabilities.
- Configured dynamic routing, inter-VLAN routing, and multi-router topologies.
- Designed enterprise-style networks incorporating routers, switches, and security appliances.
- Diagnosed complex networking issues and optimized performance through iterative testing.

Solar-Powered Embedded Control System *- Arduino · Embedded C · Energy Systems*

- Designed an off-grid control system integrating hardware, firmware, and power calculations to optimize energy usage for a solar-powered appliance.
- Produced full technical documentation, schematics, and deployment-ready impact analysis.

EDUCATION

Cape Peninsula University of Technology ***2023-2025***

Bachelor of Computer Engineering

- *Relevant Coursework: Electrical Engineering Principles I, Computer Architecture I, Electronics I, Physics I, Engineering Mathematics I, Software Design I, Engineering Ethics I, Engineering Communication I, Computer Graphics II, Signal Processing II, Digital Systems II, Operating Systems II, Engineering Mathematics II, Software Design II, Database Systems III, Electronic Communications III, Embedded Systems III, Network Systems III, Industrial Computing Design III*
- *Active participant in the tech space, contributing to circuit design, group projects, and academic presentations.*

Matric | Excelsior Comprehensive School **2022**

Achievement: Bachelor's Pass

CERTIFICATIONS & TRAINING

- FNB App Academy - Full-Stack Development (2025)
- Cisco Networking Foundations: IP Addressing
- Practical SDN & OpenFlow (LinkedIn Learning)
- Introduction to Linux

ADDITIONAL

Languages: English (Fluent), Xhosa, Zulu

Availability: Full-time from December 2025 | Open to relocation (local and international)

REFRENCES

Dr. Argus Brandt
Supervisor, CPUT Lecturer
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Mr. M. Mangquzana
Teacher, Excelsior CHS
Tel: 073 626 3414